

Roll No.

TBT-502

**B. TECH. (BIOTECHNOLOGY)
(FIFTH SEMESTER)**

MID SEMESTER EXAMINATION, Oct., 2022

BIOPROCESS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Describe thermal death kinetics of micro-organisms. What is del factor ?
10 Marks (CO1)

OR

- (b) Discuss *five* steps of Plackett Burman design with example.
10 Marks (CO1)

2. (a) What are the properties of media ? Explain effect of temperature and ionic strength of media on growth kinetics.
10 Marks (CO1)

OR

- (b) What are assumptions of Richards method ? Explain graphical method of Del estimation.
10 Marks (CO1)

P. T. O.

3. (a) *Bacillus subtilis* has a mass doubling time of 2.4 h when grown on nutrient agar. The saturation constant using this substrate is 1.3 g/l (which is unusually high) and cell yield on nutrient agar is μ_m 0.46 g cell/g nutrient agar. If we operate a chemostat on a feed stream containing 38 g/l nutrient agar a kinetic parameter value of 0.42 h^{-1} , find the following :

10 Marks (CO2)

- (i) Cell concentration
- (ii) Optimum dilution rate
- (iii) Cell productivity when dilution rate is half of the maximum

OR

- (b) Compare biomass and product concentration in Batch, Fed Batch and Continuous culture systems.

10 Marks (CO2)

4. (a) Derive an expression for total biomass concentration in Fed batch culture.

10 Marks (CO2)

OR

(3)

- (b) Discuss Monod equation and its applications. What is double reciprocal plot ?

10 Marks (CO2)

5. (a) Calculate del factor for given data using Richard's rapid calculation method :

Time (min)	Temp (°C)
14	90
16	100
18	110
20	121
22	115
24	100
26	97

Given $E/R = 34043 \text{ k}$; $k_0 = 9.5 \times 10^{37}$.

10 Marks (CO1/CO2)

OR

- (b) Discuss product profile in batch culture along with graphs applicable in each case.

10 Marks (CO1/CO2)